Ganesh Gupta

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Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Kept node routing at **sub-200ms p99** for pipeline workloads processing **millions of daily transactions**, by building a **distributed executor** in **Python** and **Java** that dispatches pipeline nodes to **Docker** or **AWS Lambda** via typed **REST APIs**.
- Sustained **99%+ uptime** and skipped recomputation on unchanged inputs, by designing a content-addressed **S3** artifact store with memoization and per-node **Redis** checkpointing so interrupted workflows resume from the last committed step.
- Cut redundant polling API calls **90%**, by replacing client polling with **server-sent events (SSE)** streaming live workflow metrics.
- Owned design to deployment of **AWS serverless** services (**Lambda, API Gateway, DynamoDB, S3**) targeting an **RTO under 2 minutes**, shipped through **GitHub Actions CI/CD** with **Codex** and **Claude Code** for **AI-assisted coding**.

QureAI | *AI Engineer* Mar 2026 – Present

- Cut **p99 latency 66% (420ms to 140ms)** across **50K+ daily requests**, by migrating synchronous model serving to **asynchronous processing** with horizontal **autoscaling** and load-aware routing.
- Reduced incident response toil, by building **CI/CD** with quality gates, failure alerting, and **automated rollback** for **safe rollouts**.
- Sped up incident triage, by building an **SRE agent** that investigates alerts and escalates data questions to a **SQL** agent.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Sustained **99.9% uptime with zero data loss** for **1M+ users**, by architecting **25+ Java Spring Boot microservices** with **REST APIs** for financial transaction processing on **AWS (EC2, RDS, SQS)**.
- Cut settlement-blocking nightly aggregation **81% (8 minutes to 90 seconds)**, by redesigning **SQL and PL/SQL** with **indexed views** and batch **partitioning** to eliminate row-level **lock contention**.
- Processed **50K+ records** per run, by designing **SQL and NoSQL** data pipelines on **AWS SQS** with **data quality** checks.
- Reduced cross-service escalations **60%** in one quarter, by standardizing retry, timeout, and failure contracts across 25+ services, leading **architecture and code reviews**, and driving **root cause analysis** and post-incident reviews.

Projects

Distributed NoSQL Database `github.com/ganeshhgupta/distributed-kv-store`

- Built a **fault-tolerant distributed database** that never confirms a write until a **majority of nodes** store it, recovers crashed nodes by replaying a **write-ahead log**, and resyncs replicas cut off by **network partitions** through **leader-follower replication**.

LangFetch `tinyurl.com/LangFetch`

- Built an agent that answers the **multi-table questions** single-shot text-to-SQL gets wrong, by breaking each question into **dependency-ordered SQL** steps that pass results forward, orchestrated with **LangGraph** and served through **FastAPI**.

RootCause AI `github.com/ganeshhgupta/rootcause-ai`

- Cut incident **MTTR 65%** with a **LangGraph multi-agent** system that triages production alerts, hands each to a specialized sub-agent with guardrailed **tool use**, carries shared state across steps, and writes a structured **root-cause brief**.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Aug 2016 – Jul 2020

Skills

Languages and Tools: Java, Python, SQL, Git, Codex, Claude Code (AI-assisted coding)

Databases: Oracle DB, PostgreSQL, DynamoDB, MongoDB, Redis, Elasticsearch, Redshift

Database Concepts: Indexing, Partitioning, Query Optimization, Replication, Quorum Consistency, Write-Ahead Logging

Backend and Distributed Systems: REST APIs, Microservices, Concurrency, Async Processing, Caching, Kafka, SQS

AWS: Lambda, API Gateway, DynamoDB, S3, SQS, RDS, EC2, Docker

Engineering Practices: DS&A, Object-Oriented Design, GitHub Actions CI/CD, Code Reviews

Certifications

AWS Certified Developer – Associate